PROBLEM 41

Rotational Kinematics

Simple Rotations, Satellite Solar Panels

Original statement · Problems.pdf, p. 29

The angle θ describing the rotation of the solar panels of a reconnaissance satellite about the body-fixed axis x is an arbitrary function of time. The satellite spins about the z axis at the constant angular rate ω. Derive expressions for the absolute velocity and acceleration of point B in the figure relative to an inertial reference frame centered in A.

Full Analytical Solution (4-Phase Methodology)

PHASE 1

Physical Statement, Hypotheses & Parameters

- **Physical System:** A reconnaissance satellite spins about its body-fixed z-axis at constant rate ω. A solar panel rotates about the body-fixed x-axis by an arbitrary prescribed angle $\theta(t)$.
- **Goal:** Determine absolute velocity $\mathbf{v}_{B/0}$ and acceleration $\mathbf{a}_{B/0}$ of tip B relative to inertial frame centered at A.
- **Geometric Parameters:** H is the distance from A to B measured along the x-axis and W is the distance from the x-axis to B (panel width), as read from the figure in Problems.pdf (PDF page 29).

PHASE 2

Reference Frames, Vector Bases & Rotation Matrices

- **Inertial Reference Frame:** $\mathcal{S}_0 = \{A; \mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$.
- **Satellite Body Frame (Precessing):** $\mathcal{S}_1 = \{A; \mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1\}$, obtained by rotating by angle $\phi = \omega t$ about $\mathbf{k}_0$:

$$[{}_0R_1] = \begin{pmatrix} \cos(\omega t) & -\sin(\omega t) & 0 \\ \sin(\omega t) & \cos(\omega t) & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \boldsymbol{\omega}_{1/0} = \omega \mathbf{k}_1$$

- **Panel-Fixed Frame:** $\mathcal{S}_2 = \{C; \mathbf{i}_2, \mathbf{j}_2, \mathbf{k}_2\}$, obtained by rotating $\mathcal{S}_1$ by angle $\theta(t)$ about $\mathbf{i}_1$:

$$[{}_1R_2] = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{pmatrix}, \quad \boldsymbol{\omega}_{2/1} = \dot{\theta}\mathbf{i}_1$$

- **Total Angular Velocity Vector (Notes Sec. 3.1, Eq. 3.14):**

$$\boldsymbol{\omega}_{2/0} = \boldsymbol{\omega}_{1/0} + \boldsymbol{\omega}_{2/1} = \dot{\theta}\mathbf{i}_1 + \omega\mathbf{k}_1$$

- **Angular Acceleration Vector:**

$$\boldsymbol{\alpha}_{2/0} = \left.\frac{d\boldsymbol{\omega}_{2/0}}{dt}\right|_{\mathcal{S}_0} = \ddot{\theta}\mathbf{i}_1 + \dot{\theta}(\boldsymbol{\omega}_{1/0} \times \mathbf{i}_1) + \dot{\omega}\mathbf{k}_1 + \omega(\boldsymbol{\omega}_{1/0} \times \mathbf{k}_1)$$

Since $\omega = \text{const}$, $\dot{\omega} = 0$ and $\boldsymbol{\omega}_{1/0} \times \mathbf{k}_1 = \mathbf{0}$.

$$\boldsymbol{\alpha}_{2/0} = \ddot{\theta}\mathbf{i}_1 + \dot{\theta}(\omega\mathbf{k}_1 \times \mathbf{i}_1) = \ddot{\theta}\mathbf{i}_1 + \omega\dot{\theta}\mathbf{j}_1$$

The hinge axis is the body-fixed x-axis ($\mathbf{i}_1$); C denotes the point of this axis at the abscissa of B, so $\mathbf{r}_{AC} = H\mathbf{i}_1$ is fixed in $\mathcal{S}_1$ and does not rotate with the panel. The axes of $\mathcal{S}_2$ expressed in $\mathcal{S}_1$ are the columns of $[{}_1R_2]$: $\mathbf{i}_2 = \mathbf{i}_1, \mathbf{j}_2 = \cos\theta\,\mathbf{j}_1 + \sin\theta\,\mathbf{k}_1, \mathbf{k}_2 = -\sin\theta\,\mathbf{j}_1 + \cos\theta\,\mathbf{k}_1$. Orthonormality check: $\det[{}_1R_2] = \cos^2\theta + \sin^2\theta = 1$ and $[{}_1R_2][{}_1R_2]^T = \text{diag}\left(1,\ \begin{pmatrix}\cos\theta & -\sin\theta \\ \sin\theta & \cos\theta\end{pmatrix}\begin{pmatrix}\cos\theta & \sin\theta \\ -\sin\theta & \cos\theta\end{pmatrix}\right) = I$.

Geometry assumption. The statement text does not fix the geometry; it is read from the figure in Problems.pdf (PDF page 29): the panel lies along the x-axis, B is the panel corner at distance H from A measured along x and at distance W from the x-axis, and θ is measured about $+\mathbf{i}_1$ from the $\mathbf{j}_1$ direction (so $\mathbf{r}_{CB} = W\mathbf{j}_2$). The zero reference of θ (panel along $+\mathbf{j}_1$ at $\theta = 0$) cannot be read unambiguously from the statement; it is the choice consistent with the official key.

- **Orthonormality of $[{}_0R_1]$:** $\det[{}_0R_1] = \cos^2\omega t + \sin^2\omega t = 1$ and $[{}_0R_1][{}_0R_1]^T = I$. Since $[{}_1R_2]$ has the same two properties, the composition $[{}_0R_2] = [{}_0R_1][{}_1R_2]$ is also a proper rotation ($\det = 1$, $RR^T = I$); all three checks were done symbolically with sympy.

PHASE 3

Step-by-Step Mathematical Deduction

Position of Point B in Basis $\mathcal{B}_1$

Decompose $\mathbf{r}_{AB} = \mathbf{r}_{AC} + \mathbf{r}_{CB}$. The first part is fixed in $\mathcal{S}_1$; the second is fixed in the panel frame $\mathcal{S}_2$ ($\mathbf{r}_{CB} = W\mathbf{j}_2$), and is brought to $\mathcal{S}_1$ with $[{}_1R_2]$:

$$\mathbf{r}_{AC} = H\mathbf{i}_1, \qquad \{\mathbf{r}_{CB}\}_1 = [{}_1R_2]\begin{pmatrix}0\\W\\0\end{pmatrix} = \begin{pmatrix}0\\W\cos\theta\\W\sin\theta\end{pmatrix}$$

$$\mathbf{r}_{AB} = H\mathbf{i}_1 + W\cos\theta\,\mathbf{j}_1 + W\sin\theta\,\mathbf{k}_1$$

Absolute Velocity of Point B

Since A is the origin of the inertial frame, $\mathbf{v}_{B/0} = \left.\frac{d\mathbf{r}_{AB}}{dt}\right|_{\mathcal{S}_0}$. By the Poisson formula, a vector fixed in $\mathcal{S}_1$ obeys $\left.\frac{d}{dt}\right|_{\mathcal{S}_0} = \boldsymbol{\omega}_{1/0}\times$ and a vector fixed in $\mathcal{S}_2$ obeys $\left.\frac{d}{dt}\right|_{\mathcal{S}_0} = \boldsymbol{\omega}_{2/0}\times$. The hub offset $H\mathbf{i}_1$ rotates only with the satellite ($\boldsymbol{\omega}_{1/0}$), not with the panel:

$$\mathbf{v}_{B/0} = \boldsymbol{\omega}_{1/0}\times\mathbf{r}_{AC} + \boldsymbol{\omega}_{2/0}\times\mathbf{r}_{CB} = \omega\mathbf{k}_1\times H\mathbf{i}_1 + (\dot\theta\mathbf{i}_1 + \omega\mathbf{k}_1)\times W(\cos\theta\,\mathbf{j}_1 + \sin\theta\,\mathbf{k}_1)$$

Cross products: $\mathbf{k}_1\times\mathbf{i}_1 = \mathbf{j}_1, \mathbf{i}_1\times\mathbf{j}_1 = \mathbf{k}_1, \mathbf{i}_1\times\mathbf{k}_1 = -\mathbf{j}_1, \mathbf{k}_1\times\mathbf{j}_1 = -\mathbf{i}_1, \mathbf{k}_1\times\mathbf{k}_1 = \mathbf{0}$. Term by term:

- $\omega \mathbf{k}_1 \times H\mathbf{i}_1 = H\omega\,\mathbf{j}_1$
- $\dot\theta \mathbf{i}_1 \times W\cos\theta\,\mathbf{j}_1 = W\dot\theta\cos\theta\,\mathbf{k}_1$
- $\dot\theta \mathbf{i}_1 \times W\sin\theta\,\mathbf{k}_1 = -W\dot\theta\sin\theta\,\mathbf{j}_1$
- $\omega \mathbf{k}_1 \times W\cos\theta\,\mathbf{j}_1 = -W\omega\cos\theta\,\mathbf{i}_1$
- $\omega \mathbf{k}_1 \times W\sin\theta\,\mathbf{k}_1 = \mathbf{0}$

Adding the five terms:

$$\mathbf{v}_{B/0} = -W\omega\cos\theta\,\mathbf{i}_1 + (H\omega - W\dot\theta\sin\theta)\,\mathbf{j}_1 + W\dot\theta\cos\theta\,\mathbf{k}_1$$

This agrees with the official key (Problems.pdf, PDF page 76).

Absolute Acceleration of Point B

Differentiate $\mathbf{v}_{B/0} = v_x\mathbf{i}_1 + v_y\mathbf{j}_1 + v_z\mathbf{k}_1$ in $\mathcal{S}_0$: the components are differentiated with the chain rule and product rule, and the Poisson formula gives $\dot{\mathbf{i}}_1 = \omega\mathbf{j}_1, \dot{\mathbf{j}}_1 = -\omega\mathbf{i}_1, \dot{\mathbf{k}}_1 = \mathbf{0}$:

$$\mathbf{a}_{B/0} = \dot v_x\mathbf{i}_1 + \dot v_y\mathbf{j}_1 + \dot v_z\mathbf{k}_1 + \omega\mathbf{k}_1 \times \mathbf{v}_{B/0} = (\dot v_x - \omega v_y)\mathbf{i}_1 + (\dot v_y + \omega v_x)\mathbf{j}_1 + \dot v_z\mathbf{k}_1$$

With $\frac{d}{dt}\cos\theta = -\dot\theta\sin\theta$ and $\frac{d}{dt}\sin\theta = \dot\theta\cos\theta$ (ω, H, W constant):

- $\dot v_x = W\omega\dot\theta\sin\theta, \quad v_y = H\omega - W\dot\theta\sin\theta, \quad \dot v_x - \omega v_y = 2W\omega\dot\theta\sin\theta - H\omega^2$
- $\dot v_y = -W(\ddot\theta\sin\theta + \dot\theta^2\cos\theta), \quad \omega v_x = -W\omega^2\cos\theta$
- $\dot v_z = W(\ddot\theta\cos\theta - \dot\theta^2\sin\theta)$

$$\mathbf{a}_{B/0} = (2W\omega\dot\theta\sin\theta - H\omega^2)\mathbf{i}_1 - (W\ddot\theta\sin\theta + W\dot\theta^2\cos\theta + W\omega^2\cos\theta)\mathbf{j}_1 + (W\ddot\theta\cos\theta - W\dot\theta^2\sin\theta)\mathbf{k}_1$$

This agrees with the official key (Problems.pdf, PDF page 76).

PHASE 4

Physical Interpretation & Dimensional Verification

- **Coriolis Coupling:** The term $2W\omega\dot{\theta}\sin\theta\,\mathbf{i}_1$ represents the Coriolis acceleration arising from the cross-coupling between satellite spin ω and panel deployment rate $\dot{\theta}$.
- **Dimensional Consistency:**
- Every velocity component has units $[\text{length}]\cdot[\text{time}]^{-1} = \text{m/s}$.
- Every acceleration component has units $[\text{length}]\cdot[\text{time}]^{-2} = \text{m/s}^2$ (Consistent).
- **Limiting cases:**
- $\dot{\theta} = 0$ (panel locked): $\mathbf{v}_{B/0} = -W\omega\cos\theta\,\mathbf{i}_1 + H\omega\,\mathbf{j}_1 = \omega\mathbf{k}_1 \times (H\mathbf{i}_1 + W\cos\theta\,\mathbf{j}_1 + W\sin\theta\,\mathbf{k}_1)$, the rigid-body velocity of a point of the spinning satellite.
- $\omega = 0$ (no spin): $\mathbf{v}_{B/0} = W\dot{\theta}(-\sin\theta\,\mathbf{j}_1 + \cos\theta\,\mathbf{k}_1)$ and the acceleration reduces to the circular-motion terms $-W\dot{\theta}^2\,\mathbf{e}_{\text{rad}} + W\ddot{\theta}\,\mathbf{e}_{\text{tan}}$ about the hinge axis ($\mathbf{e}_{\text{rad}} = \mathbf{j}_2$, $\mathbf{e}_{\text{tan}} = \mathbf{k}_2$, radius W).
- **Symbolic check (sympy):** differentiating twice in the inertial frame the vector $[{}_0R_1]\,(H,\ W\cos\theta,\ W\sin\theta)^T$ and projecting back on $\mathcal{B}_1$ with $[{}_0R_1]^T$ gives exactly the $\mathbf{v}_{B/0}$ and $\mathbf{a}_{B/0}$ of Phase 3 (all components, zero residual).

Source: Problems.pdf (Ginsberg, Cambridge Univ. Press, 2008)